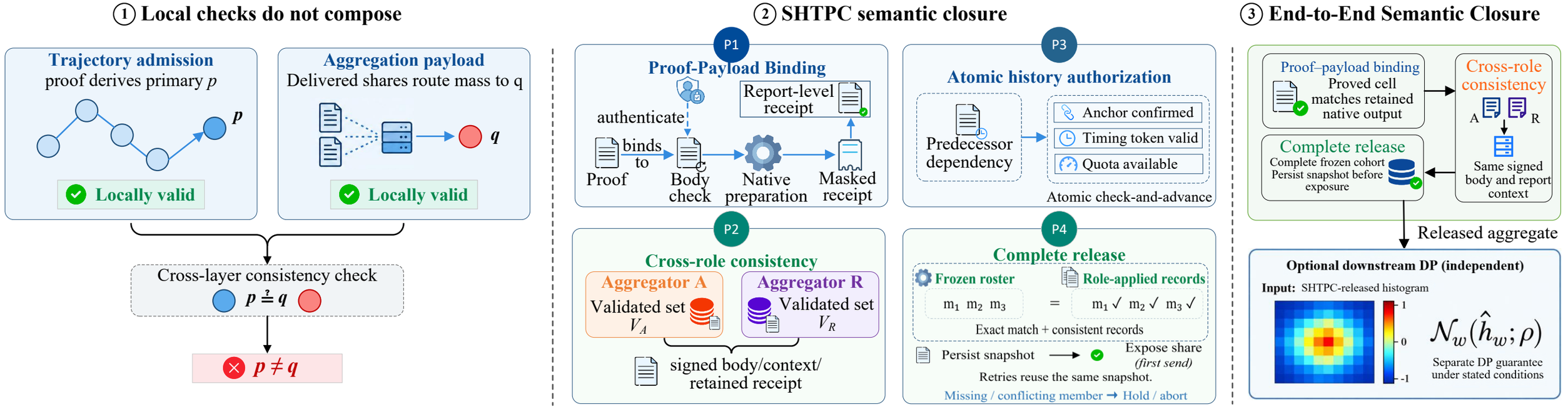

① Local checks do not compose
Trajectory admission
proof derives primary p
p
Locally valid
Aggregation payload
Delivered shares route mass to q
q
Locally valid
Cross-layer consistency check
p ≟ q
p ≠ q
② SHTPC semantic closure
P1
Proof-Payload Binding
Report-level receipt
authenticate
binds to
Proof
Body check
Native preparation
Masked receipt
P3
Atomic history authorization
Predecessor dependency
Anchor confirmed
Timing token valid
Quota available
Atomic check-and-advance
P2
Cross-role consistency
Aggregator A
Validated set
V_A
Aggregator R
Validated set
V_R
signed body/context/ retained receipt
P4
Complete release
Frozen roster
Role-applied records
m1 m2 m3
=
m1 ✓ m2 ✓ m3 ✓
Exact match + consistent records
Persist snapshot
Expose share
(first send)
Retries reuse the same snapshot.
Missing / conflicting member → Hold / abort
③ End-to-End Semantic Closure
Proof–payload binding
Proved cell matches retained native output
Cross-role consistency
A
R
Same signed body and report context
Complete release
Complete frozen cohort
Persist snapshot before exposure
Released aggregate
Optional downstream DP (independent)
Input: SHTPC-released histogram
1
0
-1
$\mathcal{N}_w(\hat{h}_w;\rho)$
Separate DP guarantee under stated conditions